

Experimental and numerical investigation of a multi-pass branching microchannel heat sink



Ercan M. Dede^{a,*}, Yan Liu^b

^a Toyota Research Institute of North America, 1555 Woodridge Ave., Ann Arbor, MI 48105, USA

^b Toyota Technical Center, 1555 Woodridge Ave., Ann Arbor, MI 48105, USA

HIGHLIGHTS

- Design and fabrication of multi-pass branching microchannel heat sink is described.
- Experiments and simulations were performed to study heat sink performance.
- Unit thermal resistance and pressure drop measurements agree with numerical results.
- Favorable thermal/flow performance achieved via multi-pass branching microchannels.
- Design approach may be used to develop novel heat transfer and fluid flow structures.

ARTICLE INFO

Article history:

Received 22 October 2012

Accepted 28 February 2013

Available online 14 March 2013

Keywords:

Heat transfer

Flow network

Stacked microchannels

ABSTRACT

This paper reports the results from an experimental and numerical investigation of a multi-pass branching microchannel heat sink for high heat flux applications. The design and fabrication of the heat sink and experimental facility is described. The unit thermal resistance and pressure drop performance of the cold plate is determined for a range of flow conditions, $1070 < Re < 6370$, using a single-phase ethylene-glycol/water coolant. Experiments are followed by local conjugate heat transfer and global flow simulations of the test piece. Comparisons are made between measured data, numerical results, and analytical predictions. The results from the investigation confirm the adequacy of the approach and reveal the effects of the multi-pass branching microchannel architecture on the performance of the cold plate. The heat sink is shown to provide favorable cooling performance with low unit thermal resistance ($0.34 \text{ cm}^2 \text{ K/W}$) and pressure drop (19.5 kPa at 1 L/min) for high-power-density (of order 100 W/cm^2) electronics applications.

© 2013 Elsevier Ltd. All rights reserved.

1. Introduction

Branching channel layouts have been shown to be beneficial in reducing the flow resistance associated with fluid networks. Constructal theory has been used to arrive at optimized duct assemblies that form tree-like structures that move from smaller to larger scales [1–3]. Fractal-like structures have also been examined and compared with parallel channel heat sinks [4], and fractal channel networks were shown to produce more uniform temperatures and significantly lower pressure drops for both disk-shaped and square-shaped flow architectures [5–8]. Additionally, heat sinks that combine jet impingement with branching channel flow have recently been investigated from both heat transfer and fluid flow

perspectives [9]. This hybrid cooling scheme was found to produce greater temperature uniformity with low pressure drop.

In separate studies, a wide variety of multi-layer heat sink designs have been shown to have potential in reducing both thermal resistance and pressure drop. The use of multiple layers is well known for conventional single-phase heat exchanger design [10], and this technique has been found to be advantageous for electronics at smaller scales both in terms of reduced surface temperatures and flow resistance [11,12]. Favorable electronics thermal performance may also be achieved using novel two-phase vapor chamber solutions; e.g. Refs. [13,14]. However, the use of vapor chambers in harsh operational environments (e.g. automotive vehicles) is still relatively limited due to tight under-hood packaging constraints, extreme ambient temperatures, and severe reliability requirements. Thus, unique straight channel single-phase heat exchange solutions [15] appear to have the greatest potential for such high heat flux (of order 100 W/cm^2) applications, although

* Corresponding author. Tel.: +1 734 995 0403; fax: +1 734 995 3053.

E-mail addresses: eric.dede@tema.toyota.com, dedeem@umich.edu (E.M. Dede).

branching channel topologies hold promise for next-generation cold plates and warrant further study as advanced manufacturing techniques grow [16,17].

Recent experimental evaluation of a single layer branching microchannel cold plate [18], designed using a structural topology optimization method, confirmed increased heat transfer with minimal change in flow resistance relative to jet impingement of a flat plate without surface enhancements. This design exploited a manifold arrangement as proposed in Refs. [19,20], where jet impingement is combined with channel flow. This concept was extended to a multi-pass branching microchannel heat sink for electronics cooling in Ref. [21], and the heat sink was again designed using numerical topology optimization. Extensive details on the optimization process for the multi-pass cold plate were provided in Ref. [21], and the influence of the thermal and fluidic optimization objective function terms, as well as mesh refinement, on the final channel flow topology was studied. Additionally, the results of purely numerical experiments comparing a single layer branching microchannel heat sink with the multi-pass design were reported in Ref. [21].

In the present work, more information is provided regarding the final fabricated multi-pass branching microchannel heat sink including details of the inlet/outlet manifolds, a newly developed test piece, and connections made to an experimental facility. Here, the experimental facility plus test piece enable accurate measurement of the real-world heat transfer and fluid flow performance of the heat sink. An in-depth description of the facility plus test section is given in this manuscript along with the experimental data collection and analysis procedure. As part of this investigation, the as-built heat sink was also physically tested over a broad range of flow rates. Comparisons are made between the measured heat sink performance, two new local-global numerical models of the experimental setup, and analytical correlations for heat transfer and pressure drop. These results indicate which modeling tools are most effective in predicting the performance of the multi-pass branching microchannel heat sink and allow for the validation of the measured unit thermal resistance and pressure drop across the novel flow structure.

In Section 2, a brief overview of the microchannel design results from the prior optimization study [21] is provided. Further details on the fabrication of the heat sink are also given in Section 2. Recent experimental work is then described in Section 3 including the test facility development, procedure, preliminary calculations, and data post-processing. The local conjugate heat transfer and global pressure drop simulations are explained in Section 4. Analytical correlations for heat transfer and pressure drop are then presented in Section 5 to supplement the experimental testing and numerical modeling. In Section 6, a heat loss analysis is provided along with

the experimental, numerical, and analytical results; an uncertainty analysis is included as a final part of this section. Conclusions are drawn in Section 7.

2. Heat sink design and fabrication

In this section, a brief overview of the multi-pass branching microchannel heat sink design is provided, and the fabrication process for the heat sink is described. The design is based on the multiphysics topology optimization studies explained in Ref. [21]. This manuscript is focused on experimental investigation of the heat sink performance along with numerical validation of the results. Thus, the optimization method is not discussed, and the reader is referred to the prior literature for information on topology optimization [22–26] plus the specific heat sink design process [18,21,27].

A two layer stacked microchannel system was used to design the heat sink. As a point of reference, the designed Layer 1 and 2 fluid flow paths from Ref. [21] are shown in Fig. 1(a) and (b), respectively. The flow paths in this figure are shown with superimposed normalized fluid velocity contours. Fluid flows through the microchannels in Layer 1 from center-to-edge, while fluid flows from edge-to-center in the Layer 2 microchannels. The full heat sink then builds off of Layers 1 and 2 and comprises five layers, where Layers 3 and 4 represent the outlet and inlet manifolds of the cold plate, respectively. Layer 5 is simply a cap that provides for tapped connections to the manifolds.

A prototype of the multi-layer heat sink was fabricated by micro-machining each plate out of aluminum and using a diffusion bonding technique to form the assembly. For reference, the average surface roughness of the microchannel walls exposed to the fluid flow was 1.6 microns based on standard precision machining tolerances. Aluminum was selected as a durable material due to its standard use in vehicle electronics cooling applications [28].

An exploded view of the bonded assembly with fluid flow path is shown in Fig. 2, where Layer 1 is on the bottom and Layer 5 is on the top. In the bonding process, the use of alloy is not required and metal atoms diffuse into the adjacent material resulting in a continuous thermal path through the thickness of the heat sink structure. The individual plates are fabricated oversize around the perimeter of the assembly and then machined to final dimensions post-bonding for a smooth exterior finish.

While [21] introduced the design of the local multi-pass region of the heat sink, Fig. 3 shows a fully finished fabricated cold plate sample, which was cross-sectioned in order to visualize the complete internal channel structure. The bonded assembly is shown in the inset image in the top left corner of the figure highlighting the aforementioned exterior surface quality. Observe that the bond

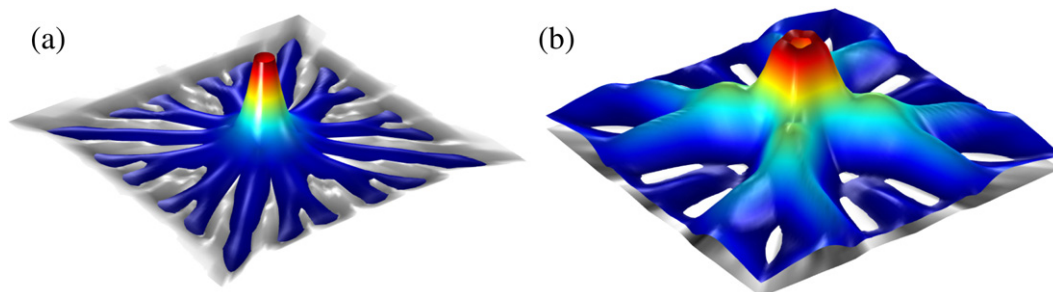


Fig. 1. Microchannel design results with superimposed normalized fluid velocity contours from Ref. [21]: (a) Layer 1 center-to-edge flow pattern; (b) Layer 2 edge-to-center flow pattern. Note: red colored contours indicate larger fluid velocity values.

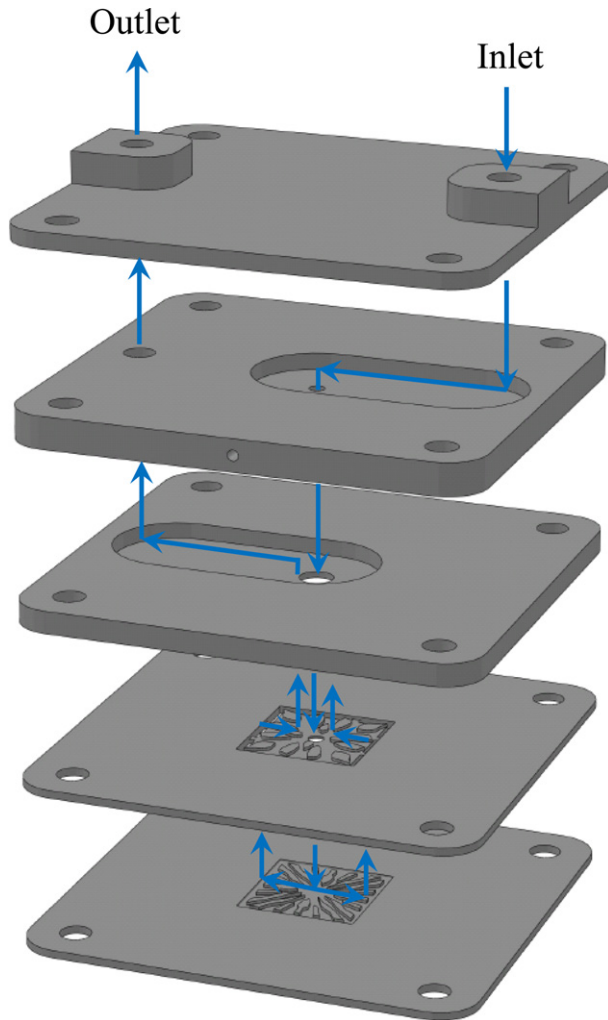


Fig. 2. Exploded view of bonded heat sink assembly showing internal manifold and microchannel structure. Layer 1 is on the bottom; Layer 5 is on the top.

lines between layers are generally not visible on the exterior or interior of the assembly. The blue arrows on the rightmost section indicate the complete coolant flow path, while the red arrows signify the heat flux region. Greater details regarding the flow path are found in Ref. [29].

The overall dimensions of the entire cold plate shown in Fig. 3 are 70 mm × 70 mm × 13.5 mm. The footprint of the local microchannel region is 20 mm × 20 mm. A large center jet orifice diameter, d , of 2.5 mm was selected to maintain low pressure drop. This jet then empties into the Layer 1 radially branching microchannel system. Layers 1 and 2 each have a final thickness of 1.25 mm with a channel depth (or fin height), H , of 0.5 mm. The inlet/outlet manifolds for the heat sink are each approximately 40 mm × 20 mm × 2.5 mm and oval in shape; see Fig. 2. The center jet orifice is constructed via a tube machined into Layer 4, which is connected to Layer 2 by the bonding process. The total orifice length is 7.34 mm including the distance traveled through Layers 3 and 4.

Based on the jet diameter and channel depth/fin height, a relatively small center jet nozzle-to-plate spacing ratio, $H/d = 0.2$, is established. The sensitivity of heat transfer and pressure drop with respect to this H/d value for a single layer branching microchannel system was previously investigated in Ref. [27]. Although a minor increase in heat transfer with lower

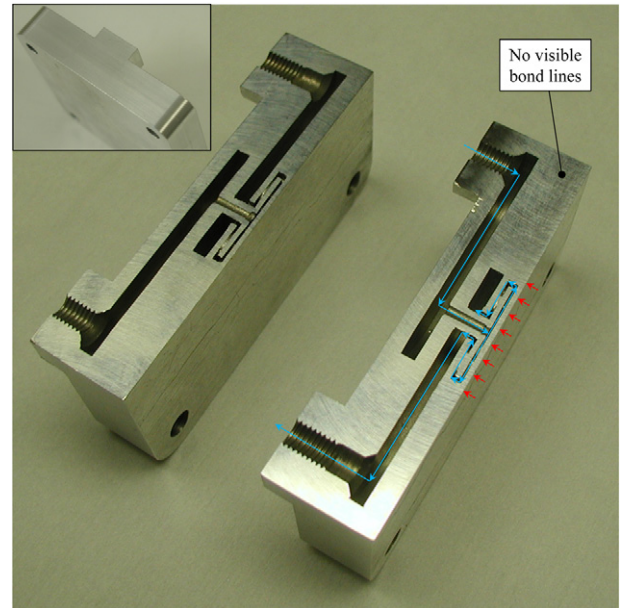


Fig. 3. Cross-section view of fabricated multi-pass branching microchannel heat sink. The fully bonded assembly prior to sectioning is shown in the image inset in the top left. The rightmost section shows the fluid flow path (blue arrows) and heat flux region (red arrows).

pressure drop is observed for larger H/d values (e.g. 0.6 and 0.8) in Ref. [27], short fins were chosen since higher aspect ratio fins create additional challenges during fabrication in terms of fin crushing due to the pressure applied during the diffusion bonding process.

3. Experimental testing

The experimental methodology is outlined in this section. The facility used to evaluate the performance of the multi-pass branching microchannel heat sink is described in Section 3.1. Details of the test piece are provided including the heat source, instrumentation, and connections to the facility. In Section 3.2, the test procedure and preliminary calculations are explained. Unit thermal resistance and pressure drop performance metrics are defined in the data post-processing procedure, Section 3.3.

3.1. Facility description

A flow loop schematic of the experimental facility is shown in Fig. 4. The facility consists of a recirculating chiller, 230 micron in-line filter, flow meter, differential pressure transducer, heater controller, inlet/outlet thermocouples, heat sink test section, and computer data acquisition system. The chiller is capable of providing steady volumetric flow rates up to 2.5 L/min with maximum system pressures of 550 kPa (80 psi). While water was used in the numerical experiments in Ref. [21], a more practical standard 50/50 ethylene-glycol/water mix was used as the coolant in this test apparatus. The fluid density, ρ , specific heat capacity, C_p , thermal conductivity, k_f , and nondimensional Prandtl number, Pr , for the fluid were approximately 1052 kg/m³, 3390 J/(kg K), 0.461 W/(m K), and 17.4, respectively.

The flow meter was a rotary style micro-flow sensor with a 0.11–2.6 L/min measurement range, accuracy of ±1% of full scale (FS), and repeatability of 0.5% FS. The differential pressure transducer had a 0–68.9 kPa measurement range with overall accuracy

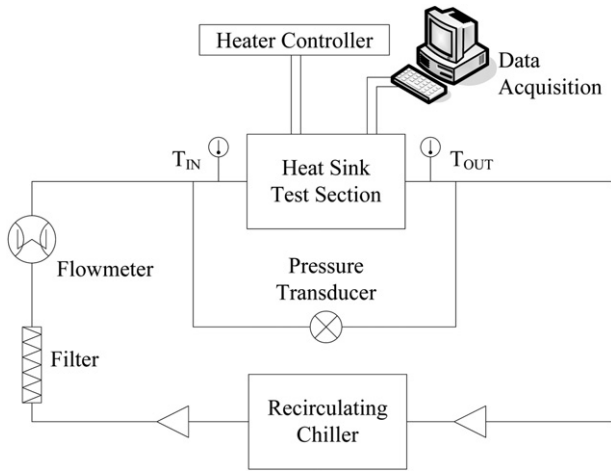


Fig. 4. Flow loop schematic of experimental facility.

of $\pm 0.25\%$ FS including repeatability of $\pm 0.05\%$ FS. Thermal span and zero effects for the pressure transducer were less than $\pm 0.02\%$ FS per degree K.

Heater power is provided to the test section via an external controller. The heater is an aluminum-nitride (AlN) ceramic device with a thermal conductivity of 140 W/(m K) , maximum allowable temperature over 423 K , 150 K/s ramp rate, and room temperature resistance of $24.9 \pm 0.1 \Omega$ (measured at 293.5 K). This device has a single internal calibrated Type K thermocouple positioned near the end of the device where the wires enter the heater. The device is capable of generating approximately 155 W/cm^2 in a package size of $25 \text{ mm} \times 15 \text{ mm} \times 2.5 \text{ mm}$. Note that the footprint area of the device is rectangular and slightly larger than the $20 \text{ mm} \times 20 \text{ mm}$ local cooling zone footprint of the cold plate described in Section 2.

Details of the test section are provided in the cross-section view in Fig. 5(a) and picture of the test setup in Fig. 5(b). Observe that the test section comprises the ceramic heater attached to the aluminum cold plate via a thermal interface material (TIM) gap pad with a thermal conductivity of 5 W/(m K) . A 25 mm thick Teflon insulation block with a 2.78 mm device cavity, not shown in Fig. 5(b) but visible in Fig. 5(a), was used to minimize device power loss to the environment and to clamp the 2.5 mm thick device to the cooler. This clamping action enforced a uniform pressure on the TIM while establishing a $0.28 \pm 0.06 \text{ mm}$ measured thickness based on part tolerances. This TIM thickness control enables the calculation of an accurate cooler surface temperature based on the recorded device temperature. Additional temperature measurements were made on the periphery of the cold plate and Teflon insulation block for numerical model calibration; see Section 4.1. Calibrated Type K thermocouples with $\pm 2.2 \text{ K}$ uncertainty were used throughout the entire test section including at the inlet/outlet flow connectors.

The custom flow test connectors were designed to interface with the cold plate inlet and outlet. Each connector provided taps for the differential pressure transducer and a thermocouple to directly measure the pressure drop across the test piece and coolant temperature. Care was taken in the design and precision machining of each tap to assure proper alignment with the flow channel and eliminate burrs, thus minimizing flow disturbances. The cross-section schematic in Fig. 5(a) shows the flow test connectors. The taps held instrument fittings (with smaller diameter fluid openings) and were located in the flow connectors just outside of the heat sink fluid flow path.

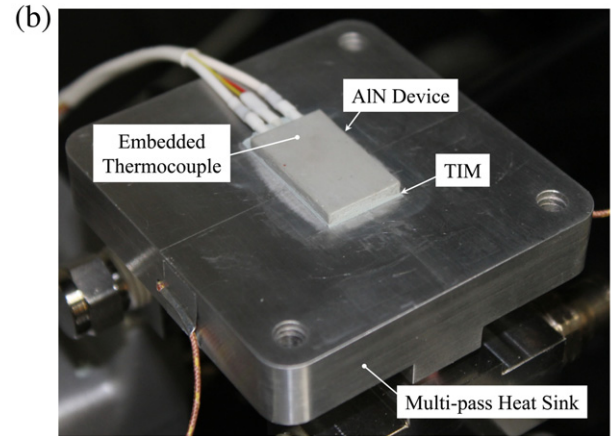
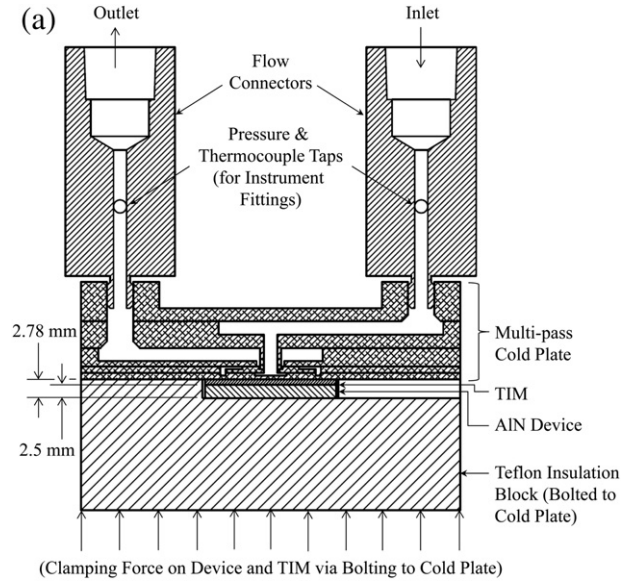


Fig. 5. Cold plate test section: (a) cross-section schematic of setup with flow test connectors installed; (b) picture of test section without insulation block.

3.2. Test procedure and preliminary calculations

The test procedure is described below along with the method for calculating the device power and heat losses to the environment. A preliminary Reynolds number analysis is provided.

A phase-angle-fire system is used to accurately control the applied heater current, which is measured via the controller using a true RMS current sensor with $\pm 0.5\%$ accuracy. Therefore, the applied power is calculated using the measured current output plus the temperature compensated heater electrical resistance. Note that a heater temperature coefficient of resistance (TCR) value of $0.00150 \pm 0.00025/\text{K}$ was used in the power calculation, per the heater vendor. Thus, the power, q_{app} , applied to the system is

$$q_{\text{app}} = I^2 R_a, \quad (1)$$

where the measured heater line current and temperature adjusted electrical resistance are given by I and R_a , respectively.

To understand the amount of power lost in the form of heat, q_{loss} , from the experimental test section to the environment, the power absorbed in the fluid, q_{abs} , may be subtracted from the applied device power [30],

$$q_{\text{loss}} = q_{\text{app}} - q_{\text{abs}} = q_{\text{app}} - \dot{m}C_p\Delta T_f, \quad (2)$$

where q_{abs} is based on the coolant mass flow rate, $\dot{m}$, specific heat capacity, C_p , and the inlet-to-outlet fluid temperature rise, ΔT_f . The power absorbed by the coolant and the measured test section temperatures may then be used to develop an accurate numerical model for post-processing of the experimental data.

As discussed in Ref. [21], the heat sink may be optimized for one set of parameters but also be subjected to a range of operating conditions. Although the heat sink was primarily designed for lower speed flows (i.e. 0.125 L/min in Ref. [21]), it was tested at much higher flow rates as reported in this work (i.e. 1 L/min, here). Thus, estimates were made of the nondimensional Reynolds number parameter, Re , within different flow regions of the test section. The commonly used Re number is given by

$$Re = \rho U D_h / \mu, \quad \text{and} \quad (3)$$

$$D_h = 4A_c/p, \quad (4)$$

where ρ and μ in Eqn. (3) are, respectively, the density and dynamic viscosity of the coolant, U is the average channel velocity, and D_h is the hydraulic diameter of the channel of interest. For the hydraulic diameter in Eqn. (4), A_c is the cross-section area of the channel, while p is the channel perimeter.

The Re numbers in three main flow regions of the test section were investigated. These regions include the inlet/outlet flow connectors ($D_h = D = 2.5$ mm for a circular channel), the jet orifice leading to the local multi-pass portion of the cold plate (again $D_h = D = 2.5$), and the branching channel with the minimum hydraulic diameter ($D_h = 0.6$ mm for a square channel) in Layer 1 of the multi-layer heat exchange portion of the structure. For the multi-pass region, the flow was assumed to be evenly split among the various radially branching channels in Layer 1. A summary of this investigation is provided in Fig. 6, where the approximate Re numbers are plotted as a function of flow rate.

Shown in Fig. 6 are two horizontal dashed lines at $Re = 2100$ and $Re = 4000$ representing the theoretical laminar flow limit and transition to turbulence, respectively, per [31]. Observe that

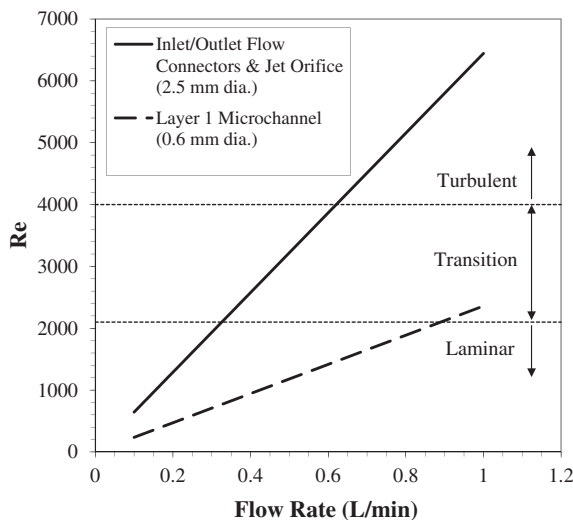


Fig. 6. Estimated Reynolds number as a function of flow rate for various flow regions of the test section.

the fluid passing through the inlet/outlet connectors and center jet orifice transitions into the turbulent regime just prior to 0.4 L/min. From this point forward, the Re number at the center jet orifice is used instead of flow rate to reveal the effects of different flow regimes on the results. Specifically, the center jet empties into the microchannel system, and the Re number in this location likely best captures the initial transition from laminar to turbulent flow within the multi-pass region of the cold plate.

Thus, experiments were conducted with center jet Re numbers spanning 1070–6370. For each test, the coolant inlet was set to an elevated temperature above 323 K. Power to the AIN device was adjusted at each flow condition to achieve a target device temperature of approximately 398 K. Since a fixed temperature gradient from device to coolant inlet was enforced, higher Re numbers corresponded to larger amounts of test piece power dissipation. The power levels achievable using the experimental apparatus were sufficient and ranged from 150 to 300 W. In all cases, adequate time was allowed for thermal/flow conditions to reach steady-state. Following this, the heat sink pressure drop was measured along with the test section temperatures over a specified time interval. Data was post-processed by taking the time average for each flow condition.

3.3. Experimental data post-processing

Following [32], it is possible to formulate the cold plate heat transfer coefficient, h_{exp} , in various ways with different temperature potentials utilized in the calculation. Thus, a local approach based on the measured heater and fluid inlet temperatures is used here

$$h_{\text{exp}} = \frac{q_{\text{app}} f_{\text{eff}}}{A_d (T_s - T_{f,\text{in}})}, \quad (5)$$

where q_{app} is the applied device power, A_d is the footprint area of the device, $T_{f,\text{in}}$ is the measured coolant inlet temperature, and T_s is the local surface temperature of the cold plate below the heater thermocouple. Inspection of the cross-section schematic in Fig. 5(a) reveals that the heat generated by the AIN device may either flow directly down into the Teflon insulation block or directly up through the TIM into the cooler. Thus, the f_{eff} term in Eqn. (5) is an efficiency factor that is defined based on the amount of power transferred as heat through the TIM into the cold plate. This efficiency factor is determined via the conjugate heat transfer numerical modeling and heat flow analysis that follow in Sections 4.1 and 6.1, respectively.

The local surface temperature, T_s , in Eqn. (5) is calculated assuming a plane wall [33] or one-dimensional (1-D) thermal circuit [34] analysis between the heater and cold plate surface by using the local measured heater thermocouple temperature, T_h , the product of the applied device power and efficiency factor, and the total thermal conduction resistance, $R_{\text{th,tot}}$, between the heater thermocouple and cooler surface

$$T_s = T_h - R_{\text{th,tot}} \cdot q_{\text{app}} \cdot f_{\text{eff}}, \quad \text{where} \quad (6)$$

$$R_{\text{th,tot}} = R_{\text{th,TIM}} + 0.5 \cdot R_{\text{th,h}}. \quad (7)$$

In Eqn. (7), $R_{\text{th,tot}}$ consists of the TIM thermal conduction resistance, $R_{\text{th,TIM}}$, plus one-half the heater thermal conduction resistance, $R_{\text{th,h}}$, since the thermocouple is embedded midway through the heater thickness. Each thermal conduction resistance term in Eqn. (7) is then evaluated using the standard relationship, per [33]

$$R_{th} = L/(k \cdot A), \quad (8)$$

where k , A , and L are the thermal conductivity, cross-section area, and thickness of the material layer of interest, respectively. The 1-D analysis assumed in extracting the local cold plate surface temperature, Eqns. (6) and (7), is based on the TIM having the same footprint as the device and the stack essentially acting as a 1-D layered composite [35]. This assumption is further examined in the context of the heat flow analysis in Section 6.1.

Given Eqns. (5)–(8), the measured ‘areal’ or unit thermal resistance, $R''_{th,exp}$, performance metric for the cold plate is then defined [33] as

$$R''_{th,exp} = \frac{1}{h_{exp}}. \quad (9)$$

As discussed in Section 3, the instrumentation taps are located in the flow connectors just outside of the heat sink fluid flow path in which various flow expansions and contractions occur. Thus, the experimentally measured pressure drop inherently captures these loss effects and is given simply by

$$\Delta P_{exp} = P_{in,exp} - P_{out,exp}, \quad (10)$$

where $P_{in,exp}$ and $P_{out,exp}$ are the respective inlet and outlet pressures measured at the taps; see Fig. 5(a).

4. Numerical modeling

Two numerical models were created to evaluate the performance of the multi-pass branching microchannel cold plate. The simulations were performed using a commercial finite element software package [36]. The first model in Section 4.1 is a local conjugate heat transfer simulation of the test piece without flow connectors. In this simulation, heat losses to the environment and test piece temperatures were matched with the measured experimental data in order to understand heat flow within the test section and to guide the post-processing of the experimental data. The second model in Section 4.2 is a global fluid flow analysis including the flow test connectors as well as the multi-pass cold plate fluid flow domain. In the latter model, the computed pressure drop was matched to the experimental results by comparing data across the same pressure measurement locations. Post-processing of the numerical data is explained in Section 4.3.

4.1. Conjugate heat transfer

A laminar flow model was utilized in the conjugate heat transfer analysis for low Re numbers (i.e. <2100), where the standard Navier–Stokes equations for the conservation of mass, momentum, and energy were solved [36,37]. For $Re > 2100$, a Reynolds-Averaged Navier–Stokes (RANS) $k-\epsilon$ turbulent flow model was selected along with the Kays–Crawford heat transport turbulence model [36]. Note that for flows close to solid boundaries and with adverse pressure gradients the $k-\epsilon$ model has some limitations in terms of predicting the spatial extent of recirculation zones [37]. Nonetheless, the model is shown to provide a reasonable trade-off between accuracy and computational cost for the test setup under consideration in this work.

A one-half symmetry model of the as-built test section was modeled, as illustrated in Fig. 7. The assumed material physical parameters of the test section components are provided in Table 1. The same thermophysical properties of the coolant described in Section 3 were used. Fluid and thermal symmetry

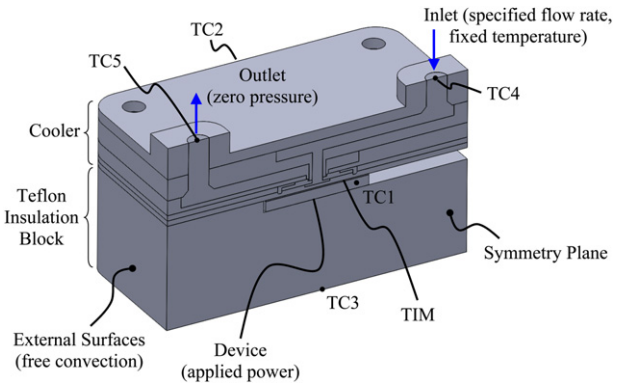


Fig. 7. Local one-half symmetry conjugate heat transfer model with boundary conditions for experimental test section. TC1–TC5 indicate locations for temperature comparison between measured and numerical data.

boundary conditions were applied to the symmetry plane defined in Fig. 7. The AlN device power calculated from the experimental data, Eqn. (1), was applied as a device load. The same coolant flow rate and elevated inlet temperature as the experiments was specified. A zero pressure boundary condition was enforced at the fluid outlet. For each simulated Re number condition, the free convection heat transfer boundary condition on the external surfaces of the test section was tuned to match the power absorbed by the coolant with the experimental value determined via Eqn. (2). Important test section temperatures were monitored at locations TC1–TC5 in Fig. 7 for calibration of the model with the experimental data.

In all simulations, the mesh density was refined to assure convergence of the solution with fluid regions assigned greater element density than solid domains. The appropriate mesh size was determined by progressively refining the mesh by approximately doubling the number of elements in the model until a convergence criterion was satisfied. Here, a change of less than 2% in the maximum temperature of the model was sought when moving from one mesh to the next. Three levels of refinement were considered with the final mesh having approximately 1.54×10^6 tetrahedral elements, where the maximum temperature for the third mesh decreased by less than 1.2% relative to the second mesh at the maximum Re number. Each conjugate heat transfer simulation required approximately 4 h on a six-core workstation with 3.46 GHz processors and 48 GB of RAM.

4.2. Pressure drop

To capture entrance and exit flow effects through the test connectors plus manifolds, and to obtain the global pressure drop across the full test section, a second fluid flow model was constructed. As described in Section 4.1, the appropriate physics were selected based on the flow Re number. For these simulations, a one-half symmetry model of the full test piece fluid flow domain was assumed including the flow connector regions. This fluid flow domain is shown in Fig. 8 and was extracted from the three-dimensional model geometry of the test section. The fluid flow rate at the inlet boundary was ramped to cover the range of

Table 1
Test section material physical parameters for the conjugate heat transfer model.

Component description	k [W/(m K)]	ρ [kg/m ³]	C_p [J/(kg K)]
Al Cooler	218	2700	895
AlN Device	140	3260	720
TIM	5	3600	1000
Teflon Insulation	0.25	2200	1050

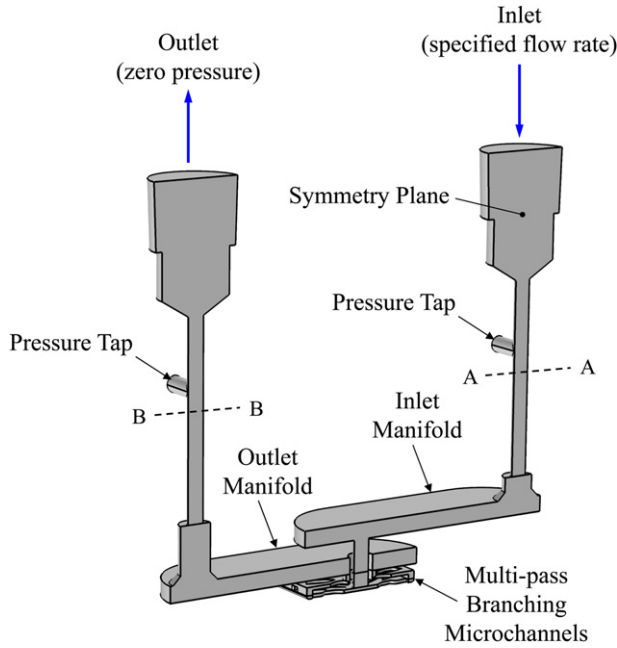


Fig. 8. Global one-half symmetry model fluid flow computational domain with boundary conditions for test piece including flow connectors plus the multi-pass branching microchannel heat sink.

experimental test conditions. A zero pressure boundary condition was enforced at the outlet.

The computational mesh was locally refined in critical regions (e.g. flow connectors and microchannels) to accurately resolve the flow field. A convergence criterion of a change of less than 2% in the simulated pressure drop (as measured across the cut lines A–A and B–B in Fig. 8) was again utilized in determining a mesh size of approximately 2.69×10^6 tetrahedral elements. The solution time for each Re number required approximately 11.5 h using the workstation described in Section 4.1.

4.3. Numerical data post-processing

A local definition similar to Eqn. (5) for the cold plate heat transfer coefficient may be utilized for the post-processing of the simulation data from the conjugate heat transfer model

$$h_{\text{sim}} = \frac{q''_{\text{local}}}{T_s - T_{f,\text{in}}} \quad (11)$$

Here, the local heat flux at the TIM to cooler interface, q''_{local} , may be post-processed directly, and this value inherently accounts for the amount of heat transferred to the cold plate via the TIM and the efficiency factor in Eqns. (5) and (6). Likewise, the local cooler surface temperature, T_s , below the heater thermocouple, TC1 in Fig. 7, may be directly obtained, and the coolant inlet temperature, $T_{f,\text{in}}$, is known *a priori*. The simulated unit thermal resistance is then

$$R''_{\text{th-sim}} = \frac{1}{h_{\text{sim}}} \quad (12)$$

For the pressure drop, Eqn. (10) is adapted

$$\Delta P_{\text{sim}} = P_{\text{in-sim}} - P_{\text{out-sim}}, \quad (13)$$

where in this case, $P_{\text{in-sim}}$ and $P_{\text{out-sim}}$, are the respective average inlet and outlet pressures computed at the cut lines A–A and B–B shown in the model in Fig. 8.

5. Analytical correlations

To supplement the measured data and numerical results, analytical correlations from the literature are proposed in Sections 5.1 and 5.2 for the cold plate unit thermal resistance and pressure drop, respectively.

5.1. Unit thermal resistance prediction

A first-order unit thermal resistance model for the heat sink allows for a comparison of the experimental, numerical, and predicted values. As suggested in Ref. [30], the predicted value, $R''_{\text{th-pre}}$, may be estimated as

$$R''_{\text{th-pre}} = \left(\frac{L}{k_{\text{al}} A_d} + \frac{1}{\eta h_{\text{pre}} A_f} + \frac{1}{2 \dot{m} C_p} \right) A_d, \quad (14)$$

where the first, second, and third terms on the right-hand-side of this expression represent the heat sink conductive, convective, and average caloric thermal resistances, respectively. In Eqn. (14), L is the heat sink base wall thickness (i.e. 0.75 mm), k_{al} is the thermal conductivity of the cooler aluminum material, A_d is the device footprint area, $\dot{m}$ is the fluid mass flow rate, and C_p is the specific heat of the coolant. The heat sink fin efficiency, η , is estimated as 0.9 based on the prior analysis in Ref. [38], and A_f is the total fin area under the device (extracted from the solid model geometry). For the predicted heat transfer coefficient, h_{pre} , the reader is referred to standard Nusselt number correlations for flat plates in parallel laminar and turbulent flow from Ref. [33], as suggested in Ref. [39]. Submerged jet impingement of very similar radial plate fin heat sinks was studied in Ref. [39], thus the same approach to heat transfer performance prediction was adopted with additional flat plates in parallel considered due to the multi-pass design. Finally, observe that lateral heat spreading effects are also not included in Eqn. (14), and this assumption is examined further in Section 6.1.

5.2. Pressure drop prediction

The analytical pressure drop for the full test section fluid flow path was calculated, per [30], with the following equation

$$\Delta P_{\text{pre}} = \Delta P_c + \Delta P_e + \Delta P_b + \Delta P_f, \quad (15)$$

where ΔP_c , ΔP_e , ΔP_b represent losses associated with contraction, expansion, and bends, respectively, and depend on the appropriate loss coefficient plus the fluid density and velocity [31]. The pressure drop due to friction, ΔP_f , through various sections of the heat sink was calculated in terms of the friction factor, channel length, coolant density, fluid velocity, and hydraulic diameter [31].

6. Results and discussion

The measured, simulated, and predicted heat sink performance is explained below. First, a heat loss analysis is provided in Section 6.1. The multi-pass branching microchannel heat sink experimental, numerical, and analytical unit thermal resistance is examined in Section 6.2 followed by the pressure drop in Section 6.3. Experimental uncertainties are also addressed in Section 6.4.

6.1. Heat loss analysis

Based on the modeling detailed in Section 4.1, the simulated and measured heater, cold plate peripheral, Teflon block, coolant inlet, and coolant outlet temperatures at locations TC1, TC2, TC3, TC4, and TC5, respectively, in Fig. 7 were matched to well within the

thermocouple measurement uncertainty (i.e. ± 2.2 K) for all flow conditions. Measured heat losses to the environment, calculated using Eqn. (2), were also matched with corresponding simulation results determined using the inlet and outlet average coolant temperatures, i.e. TC4 and TC5. The smallest amount of power was absorbed by the coolant at the lowest tested Re number with 28.2% of the applied device power lost to the environment. At the highest Re number, less than 5% of the applied power was lost to the environment.

As discussed in Section 3.3, the heat generated by the AIN device may flow through one of two paths: 1) directly into the Teflon block, or 2) through the TIM into the cooler. The amount of power transferred as heat to the cooler was obtained using the conjugate heat transfer model by integrating the heat flux at the TIM-to-cooler boundary and comparing the result with that obtained by integrating the heat flux at the device-to-Teflon block boundary. The percentage of device power transferred to the cold plate was deemed relatively insensitive to the coolant flow rate. At each Re number value, approximately 92% of the device power was transferred from the TIM to the cooler versus 8% of the dissipated power flowing into the Teflon block. Thus, an efficiency factor, $f_{eff} = 0.92$, was used in the post-processing of the experimental data, Eqns. (5) and (6).

Greater understanding of heat flow is obtained via the conjugate heat transfer model and an equivalent Biot number analysis that relates conduction in the heat sink to convective heat transfer in the fluid. Following [32,34], the dimensionless Biot number, Bi , parameter is computed as

$$Bi = \frac{h_{sim}L}{k_{al}} \quad (16)$$

where h_{sim} is the heat transfer coefficient obtained from Eqn. (11), L is the heat sink base wall thickness between the cooler exterior surface and microchannels (i.e. 0.75 mm), and k_{al} is the thermal conductivity of the heat sink aluminum material; see Table 1. Relatively small Bi numbers of 0.054 and 0.097 were obtained for low and high Re numbers, respectively, confirming that the 1-D heat conduction assumption in the experimental data post-processing, Section 3.3, and analytical prediction, Section 5.1, is reasonable. To check this result, Bi numbers were also calculated by numerically post-processing the average convective heat transfer coefficient on the actual walls of the microchannel system, and these average values were used instead of the local values for h_{sim} in Eqn. (16). A similar range of Bi numbers was obtained using this alternative approach.

6.2. Unit thermal resistance

The measured unit thermal resistance versus Re number is plotted in Fig. 9 along with the simulation results and analytical predictions. The experimental results appear repeatable with three overlapping sets of data determined using Eqns. (5)–(9). The measured data shown in Fig. 9 includes Runs 1 and 2 with $1070 < Re < 2650$ and Run 3 with $1270 < Re < 6370$. A maximum unit thermal resistance of 0.882 ± 0.255 cm² K/W was measured at $Re = 1070$ versus a minimum unit thermal resistance of 0.341 ± 0.217 cm² K/W at $Re = 6370$. The numerical data was computed using Eqns. (11) and (12), and analytical predictions are based on Eqn. (14).

Observe that the laminar ($Re < 2100$) and turbulent flow numerical results adequately predict the cold plate heat transfer for all flow conditions. As discussed in Ref. [40], for very small micro-channel dimensions, it may be difficult to sustain turbulent flows. However, for $Re > 2100$, the relatively large diameter fluid jet

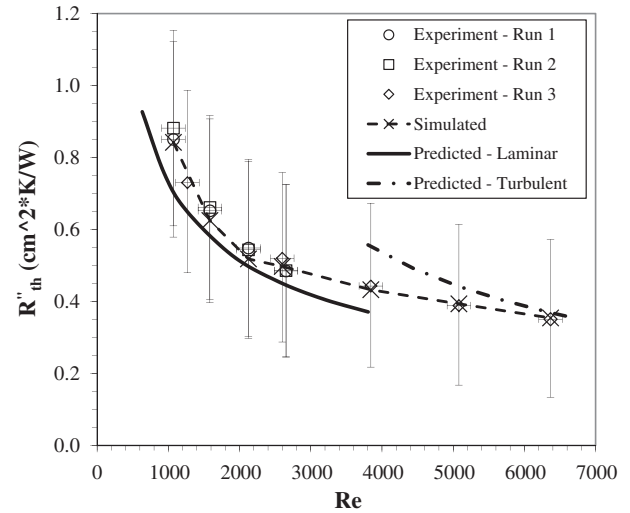


Fig. 9. Measured, simulated, and predicted unit thermal resistance versus Re number.

entering the microchannel network has transition flow characteristics, and enhanced fluid mixing was shown to occur in the region connecting the Layer 1 and 2 microchannels; see Ref. [21]. Similar transition flow conditions are likely present within the microchannels, as a result. Thus, the chosen turbulence model appears to be a reasonable selection in predicting the performance of the multi-pass microchannel region at both intermediate and larger Re numbers.

In terms of the analytical prediction, heat transfer is slightly overestimated in the laminar flow regime ($Re < 2100$), although the prediction closely follows the measured data trend. The laminar prediction was extended into the transition flow regime, $2100 < Re < 4000$, to show that the prediction begins to diverge from the experimental data at higher Re numbers. Moving to the turbulent flow heat transfer prediction, the mismatch is large at $Re = 4000$, although the predicted and experimental curves converge at the highest measured Re number.

6.3. Pressure drop

The measured pressure drop versus Re number is shown in Fig. 10(a). The results again appear repeatable with a pressure drop of 1.6 ± 1.36 kPa measured at $Re = 1070$. For $Re = 6370$, a pressure drop of 35.7 ± 3.15 kPa was measured across the instrumentation taps. The simulation results and analytical predictions, also shown in Fig. 10(a), are in good agreement with the experimental data, which allows for a detailed breakdown of the pressure drop throughout the test section. Note that the laminar flow prediction was extended up to $Re = 4000$, although the turbulent flow prediction matches the measured data for $2100 < Re < 4000$, as well. Fig. 10(b) shows a comparison of the computed and predicted pressure as a function of distance traveled through the test piece at $Re = 6370$. Inspection of both images in Fig. 10 reveals that the jet orifice and multi-pass branching microchannel portions of the heat sink account for approximately 55% of the total computed pressure drop (~ 19.5 kPa at $Re = 6370$) with the inlet/outlet connectors and manifolds responsible for the remainder; see further details in Ref. [29].

6.4. Experimental uncertainties

Shown in Fig. 9, a careful heat transfer uncertainty analysis provides the vertical error bars ranging from $\pm 29\%$ at $Re = 1070$ to $\pm 64\%$ at $Re = 6370$. Here, the thermocouple temperature (± 2.2 K),

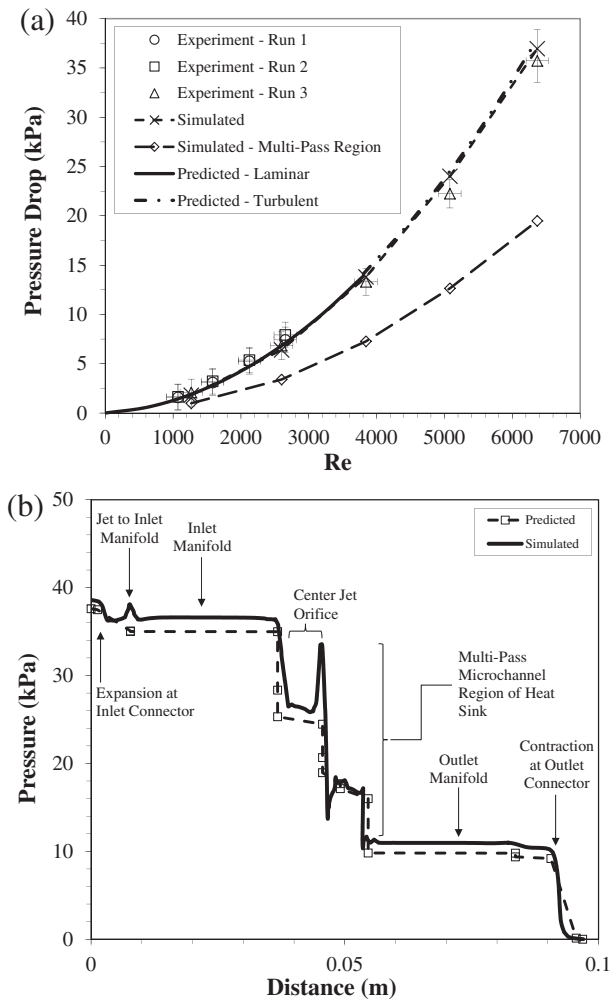


Fig. 10. Pressure drop results: (a) measured, simulated, and predicted pressure drop versus Re number; (b) simulated and predicted pressure versus distance traveled through the test piece at $Re = 6370$.

TIM thickness (± 0.06 mm), device resistance (± 0.1 Ω), device TCR (± 0.00025 /K), and device current ($\pm 0.5\%$) errors were accounted for in this analysis. The small horizontal error bars in Fig. 9 represent the Re number uncertainty due to the micro-flow sensor ($\pm 1\%$ FS).

For the pressure drop uncertainty analysis, the vertical error bars in Fig. 10(a) range from $\pm 65\%$ at $Re = 1070$ to $\pm 8\%$ at $Re = 6370$. Sources of error include pressure measurement variations (± 0.079 to ± 1.88 kPa), transducer accuracy ($\pm 0.25\%$ FS), and transducer thermal span and zero effects ($\pm 0.02\%$ FS). The horizontal error bars in Fig. 10 are also small and are again related to the accuracy of the flow meter.

Overall, the numerical and predicted results in Figs. 9 and 10 fall within the range of uncertainty for the measured data. These results are satisfactory considering the complexity of the designed fluid flow path. This suggests that the analytical approach detailed in this paper is a reasonable first-order approximation of the cold plate characteristics, while the numerical strategy logically provides greater in-depth understanding of the heat sink performance over multiple flow regimes.

7. Conclusions

The design and fabrication of a multi-pass branching micro-channel heat sink for electronics cooling was briefly reviewed.

Experimental measurements were carried out over a broad range of flow rates to understand the thermal and flow performance of the as-built design. Numerical and analytical tools were introduced for performance prediction and validation of experimental results. Unit thermal resistance and pressure drop metrics for the cold plate were quantified and good agreement was found between the results from the various analysis approaches. Generally, favorable single-phase heat transfer ($R''_{th} = 0.34$ cm² K/W) with low pressure drop (19.5 kPa at 1 L/min) for high heat fluxes (of the order 100 W/cm²) is achievable utilizing the multi-pass branching microchannel cold plate concept. Logical extensions of this design include configuring an array of local cooling cells with optimized global inlet and outlet manifolds for the thermal management of multiple devices. The design approach may be exploited in the development of a variety of novel heat transfer and fluid flow structures.

References

- [1] A. Bejan, Constructal tree network for fluid flow between a finite-size volume and one source or sink, *Rev. Gen. Therm.* 36 (1997) 592–604.
- [2] A. Bejan, S. Lorente, Constructal tree-shaped flow structures, *Appl. Therm. Eng.* 27 (2007) 755–761.
- [3] L.A.O. Rocha, S. Lorente, A. Bejan, Tree-shaped vascular wall designs for localized intense cooling, *Int. J. Heat Mass Transfer* 52 (2009) 4535–4544.
- [4] D. Pence, Reduced pumping power and wall temperature in microchannel heat sinks with fractal-like branching channel networks, *Microscale Thermophys. Eng.* 6 (2002) 319–330.
- [5] X.-Q. Wang, P. Xu, A.S. Mujumdar, C. Yap, Thermal characteristics of tree-shaped microchannel nets for cooling of a rectangular heat sink, *Int. J. Therm. Sci.* 45 (2006) 1103–1112.
- [6] D. Pence, The simplicity of fractal-like flow networks for effective heat and mass transport, *Exp. Therm. Fluid Sci.* 34 (2010) 474–486.
- [7] X.-Q. Wang, P. Xu, A.S. Mujumdar, C. Yap, Flow and thermal characteristics of offset branching network, *Int. J. Therm. Sci.* 49 (2010) 272–280.
- [8] Y. Chen, P. Cheng, Heat transfer and pressure drop in fractal tree-like micro-channel nets, *Int. J. Heat Mass Transfer* 45 (2002) 2643–2648.
- [9] J. Barrau, D. Chemisana, J. Rosell, L. Tadrist, M. Ibanez, An experimental study of a new hybrid jet impingement/micro-channel cooling scheme, *Appl. Therm. Eng.* 30 (2010) 2058–2066.
- [10] W.M. Kays, A.L. London, *Compact Heat Exchangers*, third ed., McGraw-Hill, New York, 1984.
- [11] K. Vafai, L. Zhu, Analysis of two-layered micro-channel heat sink concept in electronic cooling, *Int. J. Heat Mass Transfer* 42 (1999) 2287–2297.
- [12] X. Wei, Y. Joshi, Stacked microchannel heat sinks for liquid cooling of microelectronic components, *J. Electron. Packag.* 126 (2004) 60–66.
- [13] A. Bar-Cohen, G. Sherwood, M. Hodes, G. Solbreken, Gas-assisted evaporative cooling of high density electronic modules, *IEEE Trans. Compon. Packag. Manuf. Technol. Part A* 18 (1995) 502–509.
- [14] S.-C. Wong, K.-C. Hsieh, J.-D. Wu, W.-L. Han, A novel vapor chamber and its performance, *Int. J. Heat Mass Transfer* 53 (2010) 2377–2384.
- [15] R.J. Kee, et al., The design, fabrication, and evaluation of a ceramic counter-flow microchannel heat exchanger, *Appl. Therm. Eng.* 31 (2011) 2004–2012.
- [16] J.P. Calame, D. Park, R. Bass, R.E. Myers, P.N. Safier, Investigation of hierarchically branched-microchannel coolers fabricated by deep reactive ion etching for electronics cooling applications, *J. Heat Transfer* 131 (2009) 051401-1–051401-9.
- [17] A. Ortega, K.S.H. Potluri, B. Hassel, An investigation of multi-layer mini-channel heat sinks with channel geometric scale variation suggested by constructal scaling principles, in: *Proceedings of the 27th Annual IEEE Semiconductor Thermal Measurement and Management Symposium (SEMI-THERM)*, 2011.
- [18] E.M. Dede, Experimental investigation of the thermal performance of a manifold hierarchical microchannel cold plate, in: *Proceedings of the ASME Pacific Rim Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Systems (InterPACK)*, 2011.
- [19] G.M. Harpole, J.E. Eninger, Micro-channel heat exchanger optimization, in: *Proceedings of the 7th IEEE SEMI-THERM Symposium*, 1991.
- [20] Y.I. Kim, J.E. Chun, J.T. Kim, B.C. Pak, B.J. Baek, Forced air cooling by using manifold microchannel heat sinks, *KSME Int. J.* 12 (1998) 709–718.
- [21] E.M. Dede, Optimization and design of a multi-pass branching microchannel heat sink for electronics cooling, *J. Electron. Packag.* 134 (2012) 041001-1–041001-10.
- [22] M.P. Bendsoe, O. Sigmund, *Topology Optimization – Theory, Methods and Applications*, Springer Verlag, Berlin, 2003.
- [23] T. Borvall, J. Petersson, Topology optimization of fluids in Stokes flow, *Int. J. Numer. Methods Fluids* 41 (2003) 77–107.
- [24] L.H. Olesen, F. Okkels, H. Bruus, A high-level programming-language implementation of topology optimization applied to steady-state Navier-Stokes flow, *Int. J. Numer. Methods Eng.* 65 (2006) 975–1001.

- [25] G.H. Yoon, Topological design of heat dissipating structure with forced convective heat transfer, *J. Mech. Sci. Technol.* 24 (2010) 1225–1233.
- [26] G.H. Yoon, Topological layout design of electro-fluid-thermal-compliant actuator, *Comput. Method Appl. Mech. Eng.* 209–212 (2012) 28–44.
- [27] E.M. Dede, The influence of channel aspect ratio on the performance of optimized thermal-fluid structures, in: *Proceedings of the COMSOL Conference*, 2010.
- [28] N. Nozawa, T. Maekawa, S. Nozawa, K. Asakura, Development of power control unit for compact-class vehicle, in: *Proceedings of the SAE World Congress & Exhibition*, 2009.
- [29] E.M. Dede, Investigation of liquid flow in a multi-pass branching micro-channel system, in: *Proceedings of the ASME International Mechanical Engineering Congress and Exhibition (IMECE)*, 2012.
- [30] D.J. Sharar, B. Morgan, N.R. Jankowski, A. Bar-Cohen, Two-phase minichannel cold plate for army vehicle power electronics, in: *Proceedings of the ASME Pacific Rim Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Systems (InterPACK)*, 2011.
- [31] B.R. Munson, D.F. Young, T.H. Okiishi, *Fundamentals of Fluid Mechanics*, fifth ed., John Wiley & Sons, Inc., Hoboken, 2006.
- [32] R. Wälchli, T. Brunschweiler, B. Michel, D. Poulikakos, Combined local microchannel-scale CFD modeling and global chip scale network modeling for electronics cooling design, *Int. J. Heat Mass Transfer* 53 (2010) 1004–1014.
- [33] F.P. Incropera, *Liquid Cooling of Electronics Devices by Single-phase Convection*, John Wiley & Sons, New York, 1999.
- [34] F.P. Incropera, D.P. Dewitt, T.L. Bergman, A.S. Lavine, *Introduction to Heat Transfer*, fifth ed., (2007). Hoboken.
- [35] M. Kaviany, *Principles of Heat Transfer*, John Wiley & Sons, New York, 2001.
- [36] COMSOL AB, *COMSOL Multiphysics Ver. 4.2a* (2011). Stockholm.
- [37] J.N. Reddy, D.K. Gartling, *The Finite Element Method in Heat Transfer and Fluid Dynamics*, second ed., CRC Press, Boca Raton, 2000.
- [38] E.M. Dede, Y. Liu, Scale effects on thermal-fluid performance of optimized hierarchical structures, in: *Proceedings of the ASME/JSM 8th Thermal Engineering Joint Conference (AJTEC)*, 2011.
- [39] P.F. Sullivan, S. Ramadhyani, F.P. Incropera, Extended surfaces to enhance impingement cooling with single circular round jets, in: *Proceedings of the ASME/JSM Conference on Electronic Packaging*, 1992.
- [40] D. Liu, S.V. Garimella, Investigation of liquid flow in microchannels, *J. Thermophys. Heat Transfer* 18 (2004) 65–72.